

EEE 431 – Digital Communications

MATLAB Assignment 2

Modulation and Demodulation

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1 Introduction

In this assignment, we study a quaternary digital modulation scheme which is given to us such that the four possible bit pairs are mapped to four passband signals with different carrier phases.

The transmitted signal during each symbol interval is :

$$s_i(t) = Ah(t) \cos \left(2\pi f_c t + (i-1) \frac{\pi}{2} \right), \quad i = 1, 2, 3, 4,$$

where $h(t) = 1$ for $0 \leq t < T$ and zero otherwise. The carrier frequency is $f_c = 20$ Hz. In the waveform generation part, $T = 0.1$ s and $A = 1$ are used.

The assignment contains both analytical and simulation based parts. First, the modulated waveform is generated. Then, the signal-space representation is obtained, and the optimum correlator receiver is derived. For equal prior probabilities, maximum likelihood (ML) detection is used. For unequal prior probabilities, maximum a posteriori (MAP) detection is used. Finally, the effect of Gray encoding on bit error probability is investigated. Therefore the assignment itself consist of what we learned in signal space and optimum receiver ideas.

2 Outline

The report is organized as follows. Section 3 presents the time-domain waveform generation. Section 4 derives the signal-space representation and the orthonormal basis functions. Section 5 gives the ML receiver and the theoretical symbol error probability. Sections 6 and 7 present Monte Carlo simulations for equal and unequal prior probabilities. Section 8 derives the relation between γ_s and γ_b . Section 9 studies the bit error probability with original and Gray mappings.

3 Part (a): Generation of the Modulated Signal

The bit-to-symbol mapping is

$$00 \rightarrow s_1(t), \quad 01 \rightarrow s_2(t), \quad 10 \rightarrow s_3(t), \quad 11 \rightarrow s_4(t).$$

In the MATLAB simulation, 10 random bits were generated as using seed "431" are

$$1011010100.$$

Grouping the bits into pairs gives

$$10, 11, 01, 01, 00.$$

Therefore, the transmitted symbol sequence is

$$s_3, s_4, s_2, s_2, s_1.$$

The symbol duration is $T = 0.1$ s, so 5 transmitted symbols occupy a total time interval of 0.5 s. Since $f_s = 2000$ Hz, the sampling period is $T_s = 1/f_s = 0.0005$ s and each symbol interval contains $Tf_s = 200$ samples.

The transmitted signal is constructed as using the formula

$$x(t) = \sum_{k=0}^{N_s-1} s_{m_k}(t - kT),$$

where $m_k \in \{1, 2, 3, 4\}$ is the symbol index during the k -th symbol interval. The generated waveform is shown in Fig. 1. The discontinuities at symbol boundaries are what I have expected since the carrier phase can change from one symbol to another.

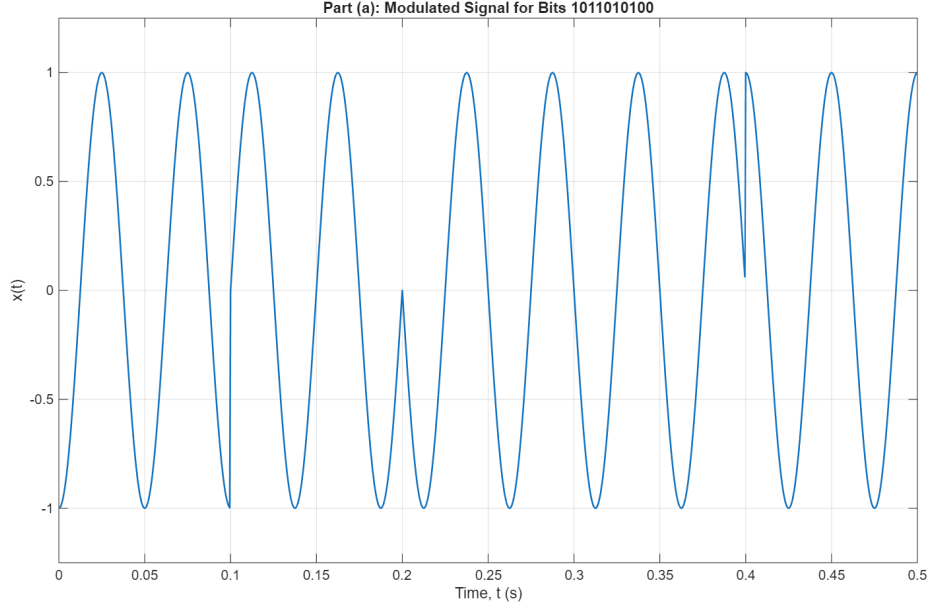


Figure 1: Generated modulated signal $x(t)$ for the random bit sequence 1011010100.

4 Part (b): Signal-Space Representation

Using the identity

$$\cos(\omega_c t + \theta_i) = \cos \theta_i \cos(\omega_c t) - \sin \theta_i \sin(\omega_c t),$$

each signal can be written as a linear combination of cosine and sine carrier components. Since

$$f_c T = 20(0.1) = 2,$$

the carrier completes exactly two cycles in one symbol interval. Therefore, the cosine and sine functions are orthogonal over $0 \leq t < T$.

A convenient orthonormal basis we use is

$$\phi_1(t) = \sqrt{\frac{2}{T}} h(t) \cos(2\pi f_c t), \quad \phi_2(t) = -\sqrt{\frac{2}{T}} h(t) \sin(2\pi f_c t).$$

The minus sign in $\phi_2(t)$ is to make the numbers simpler.

After I have coded this in the MATLAB and did the orthonormality check from MATLAB gave

$$\langle \phi_1, \phi_1 \rangle = 1.000000, \quad \langle \phi_2, \phi_2 \rangle = 1.000000, \quad \langle \phi_1, \phi_2 \rangle = 9.880985 \times 10^{-18}.$$

Thus, the two basis functions are orthonormal. The signal-space dimension is

$$\boxed{2}.$$

The basis functions are plotted in Fig. 2.

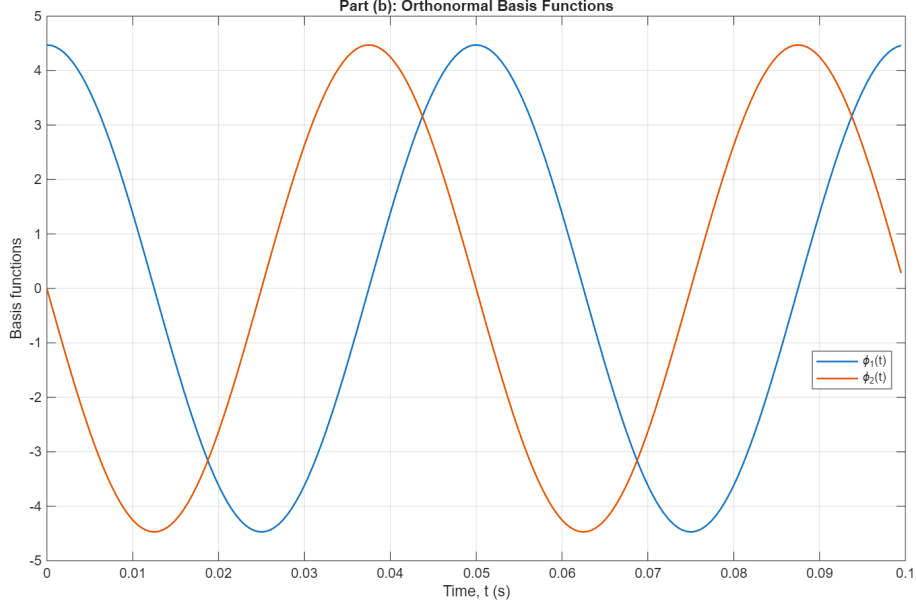


Figure 2: The orthonormal basis functions $\phi_1(t)$ and $\phi_2(t)$ over one symbol interval.

Let

$$\theta_i = (i - 1) \frac{\pi}{2}.$$

Then

$$s_i(t) = A \sqrt{\frac{T}{2}} \cos \theta_i \phi_1(t) + A \sqrt{\frac{T}{2}} \sin \theta_i \phi_2(t).$$

Define

$$a = A \sqrt{\frac{T}{2}}.$$

The four signals are represented by the vectors

$$\mathbf{s}_1 = (a, 0), \quad \mathbf{s}_2 = (0, a), \quad \mathbf{s}_3 = (-a, 0), \quad \mathbf{s}_4 = (0, -a).$$

The energy of each signal is

$$E_s = \|\mathbf{s}_i\|^2 = a^2 = \frac{A^2 T}{2}.$$

Therefore, to make each signal have energy E_s , the amplitude must be

$$A = \sqrt{\frac{2E_s}{T}}.$$

For the waveform-generation value $A = 1$, the symbol energy is

$$E_s = \frac{1^2(0.1)}{2} = 0.05,$$

and therefore $a = \sqrt{0.05} = 0.2236$. The corresponding signal vectors are

$$\mathbf{s}_1 = (0.2236, 0), \quad \mathbf{s}_2 = (0, 0.2236), \quad \mathbf{s}_3 = (-0.2236, 0), \quad \mathbf{s}_4 = (0, -0.2236).$$

The constellation is shown in Fig. 3.

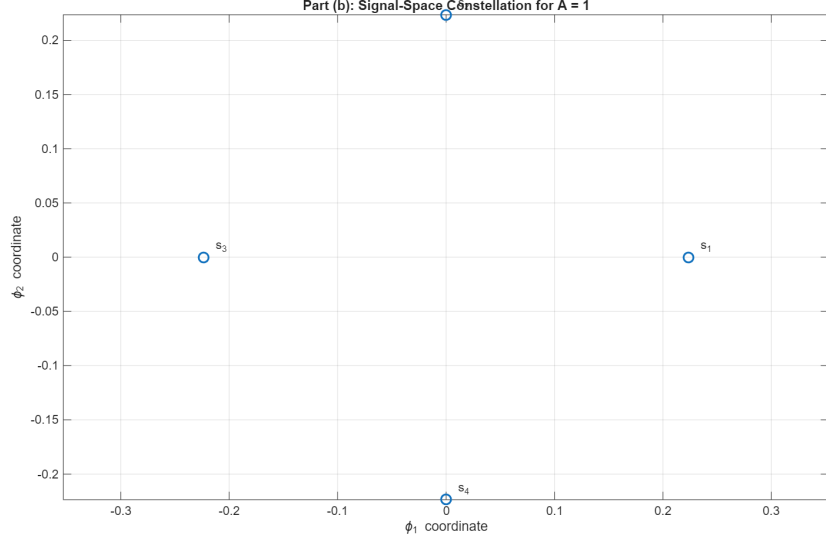


Figure 3: Signal-space constellation for $A = 1$. The four points lie on the coordinate axes.

5 Part (c): ML Receiver for Equal Prior Probabilities

Since the transmission is over an AWGN channel. The received signal is

$$r(t) = s_i(t) + n(t),$$

where $n(t)$ is white Gaussian noise with two-sided power spectral density $N_0/2$. The correlator receiver projects $r(t)$ onto the two basis functions:

$$r_1 = \int_0^T r(t) \phi_1(t) dt, \quad r_2 = \int_0^T r(t) \phi_2(t) dt.$$

Thus, the receiver obtains the vector

$$\mathbf{r} = (r_1, r_2) = \mathbf{s}_i + \mathbf{n},$$

where

$$n_1, n_2 \sim \mathcal{N}\left(0, \frac{N_0}{2}\right)$$

and the noise components are independent.

The correlator receiver structure is summarized as follows:

$$r(t) \longrightarrow \begin{cases} r_1 = \int_0^T r(t) \phi_1(t) dt, \\ r_2 = \int_0^T r(t) \phi_2(t) dt \end{cases} \longrightarrow (r_1, r_2) \longrightarrow \text{Decision Device} \longrightarrow \hat{s}_i$$

Figure 4: Optimal correlator-based receiver for the two-dimensional signal space.

Since the input bits are equally likely, all four symbols are equally likely. Therefore, the optimal decision rule is the ML rule:

$$\hat{i} = \arg \max_i f(\mathbf{r} | \mathbf{s}_i).$$

For AWGN, this is equivalent to nearest-neighbor detection:

$$\boxed{\hat{i} = \arg \min_i \|\mathbf{r} - \mathbf{s}_i\|^2.}$$

Since all four signal energies are equal, this can also be written as

$$\boxed{\hat{i} = \arg \max_i \mathbf{r} \cdot \mathbf{s}_i.}$$

5.1 Theoretical Symbol Error Probability

For the normalized constellation, let's take $\mathbf{s}_1 = (\sqrt{E_s}, 0)$. By symmetry, all symbols have the same probability of error. The correct decision region for $\mathbf{s}_1$ is the wedge where the received point is closer to $\mathbf{s}_1$ than to the other three points. This gives the two conditions

$$\sqrt{E_s} + n_1 > n_2, \quad \sqrt{E_s} + n_1 > -n_2.$$

Equivalently,

$$n_1 - n_2 > -\sqrt{E_s}, \quad n_1 + n_2 > -\sqrt{E_s}.$$

The two random variables $n_1 - n_2$ and $n_1 + n_2$ are independent Gaussian random variables with variance N_0 . Therefore, the probability of correct decision is

$$P_c = \left[1 - Q\left(\sqrt{\frac{E_s}{N_0}}\right) \right]^2 = [1 - Q(\sqrt{\gamma_s})]^2.$$

Hence, the symbol error probability is

$$P_s = 1 - P_c = 1 - [1 - Q(\sqrt{\gamma_s})]^2.$$

Therefore,

$$\boxed{P_s = 2Q(\sqrt{\gamma_s}) - Q^2(\sqrt{\gamma_s})}$$

where

$$\gamma_s = \frac{E_s}{N_0}.$$

In the MATLAB implementation, the Q -function was evaluated using

$$Q(x) = \frac{1}{2} \operatorname{erfc}\left(\frac{x}{\sqrt{2}}\right).$$

6 Parts (d)–(e): Equal-Prior ML Monte Carlo Simulation

The ML receiver was simulated using 10^6 transmitted symbols in this experiment. Instead of generating noisy passband waveforms for all symbols, the simulation was performed directly in the two-dimensional signal space. This is valid because the correlator outputs (r_1, r_2) are sufficient statistics for ML detection in AWGN.

For the simulation, the constellation was normalized to $E_s = 1$:

$$\mathbf{s}_1 = (1, 0), \quad \mathbf{s}_2 = (0, 1), \quad \mathbf{s}_3 = (-1, 0), \quad \mathbf{s}_4 = (0, -1).$$

For each value of γ_s , the noise variance was selected from

$$N_0 = \frac{E_s}{\gamma_s}, \quad \sigma^2 = \frac{N_0}{2}.$$

The simulated and theoretical symbol error probabilities are given in Table 1.

Table 1: Equal-prior ML symbol error probability.

γ_s (dB)	Theoretical P_s	Simulated P_s
-8	5.713360×10^{-1}	5.714920×10^{-1}
-6	5.213016×10^{-1}	5.214900×10^{-1}
-4	4.583544×10^{-1}	4.590930×10^{-1}
-2	3.814212×10^{-1}	3.813130×10^{-1}
0	2.921390×10^{-1}	2.929140×10^{-1}
2	1.972353×10^{-1}	1.970510×10^{-1}
4	1.097989×10^{-1}	1.099430×10^{-1}
6	4.548495×10^{-2}	4.537200×10^{-2}
8	1.197272×10^{-2}	1.205000×10^{-2}
10	1.564790×10^{-3}	1.558000×10^{-3}
12	6.860407×10^{-5}	5.900000×10^{-5}

The curves are shown in Fig. 5. The simulation follows the theoretical expression very closely. At high γ_s , small deviations are expected because the number of observed errors becomes small. For example, when the error probability is on the order of 10^{-5} , only several tens of errors are observed in 10^6 symbols.

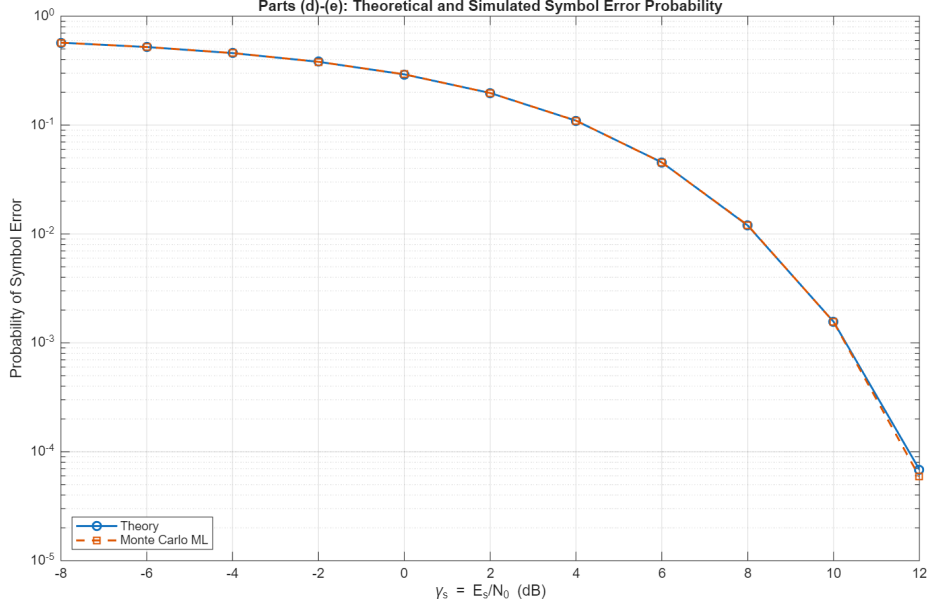


Figure 5: Theoretical and simulated symbol error probabilities for the equal-prior ML receiver.

7 Part (f): MAP Receiver for Unequal Prior Probabilities

Now in this experiment the symbols are not equally likely. The prior probabilities are

$$P_1 = 0.1, \quad P_2 = 0.2, \quad P_3 = 0.3, \quad P_4 = 0.4.$$

The optimal receiver is no longer the ML receiver. Instead, the optimal receiver that minimizes the average symbol error probability is the MAP receiver:

$$\hat{i} = \arg \max_i P_i f(\mathbf{r} | \mathbf{s}_i).$$

For AWGN,

$$f(\mathbf{r} | \mathbf{s}_i) \propto \exp\left(-\frac{\|\mathbf{r} - \mathbf{s}_i\|^2}{N_0}\right).$$

Therefore, the MAP decision rule is

$$\hat{i} = \arg \max_i \left[\ln P_i - \frac{\|\mathbf{r} - \mathbf{s}_i\|^2}{N_0} \right].$$

Equivalently,

$$\hat{i} = \arg \min_i \left[\|\mathbf{r} - \mathbf{s}_i\|^2 - N_0 \ln P_i \right].$$

Since all signal energies are equal, this can also be written as

$$\hat{i} = \arg \max_i \left[\mathbf{r} \cdot \mathbf{s}_i + \frac{N_0}{2} \ln P_i \right].$$

The ML receiver from Part (c) ignores the prior probabilities. Therefore, it remains optimal only when all symbols are equally likely. For the unequal prior probabilities in this part, the ML receiver is not optimal; the MAP receiver is optimal.

8 Parts (g)–(h): MAP and ML Simulation under Unequal Priors

In this part, the symbol sequence was generated according to the probabilities

$$[0.1, 0.2, 0.3, 0.4].$$

The MAP receiver used the metric

$$\Lambda_i^{\text{MAP}}(\mathbf{r}) = \mathbf{r} \cdot \mathbf{s}_i + \frac{N_0}{2} \ln P_i,$$

whereas the ML receiver used the metric

$$\Lambda_i^{\text{ML}}(\mathbf{r}) = \mathbf{r} \cdot \mathbf{s}_i.$$

The simulation results are given in Table 2.

Table 2: MAP and ML symbol error probabilities under unequal prior probabilities.

γ_s (dB)	MAP receiver P_s	ML receiver P_s
-8	5.077650×10^{-1}	5.709470×10^{-1}
-6	4.668780×10^{-1}	5.219330×10^{-1}
-4	4.137360×10^{-1}	4.595060×10^{-1}
-2	3.465190×10^{-1}	3.824760×10^{-1}
0	2.656750×10^{-1}	2.918860×10^{-1}
2	1.798270×10^{-1}	1.964880×10^{-1}
4	1.003700×10^{-1}	1.095550×10^{-1}
6	4.178600×10^{-2}	4.540200×10^{-2}
8	1.093200×10^{-2}	1.195000×10^{-2}
10	1.526000×10^{-3}	1.647000×10^{-3}
12	6.100000×10^{-5}	7.600000×10^{-5}

The comparison is shown in Fig. 6. The MAP receiver has lower error probability than the ML receiver for the unequal-prior case. This is expected because MAP uses both the received vector and the prior probabilities. The improvement is more visible at low and moderate SNR values. At high SNR, the received vector is usually close to the transmitted constellation point, so the prior correction becomes less important and the two receivers become closer.

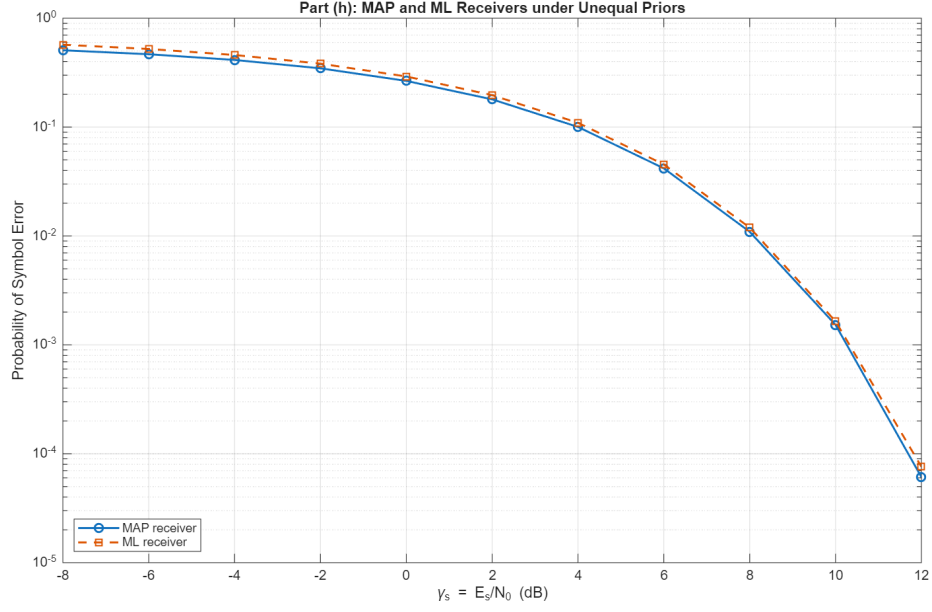


Figure 6: Comparison of MAP and ML receivers under unequal prior probabilities.

9 Part (i): Relation between γ_s and γ_b

The modulation is quaternary, so

$$M = 4.$$

Each transmitted symbol carries

$$\log_2 M = \log_2 4 = 2$$

bits. Therefore, the symbol energy and bit energy satisfy

$$E_s = 2E_b.$$

Dividing both sides by N_0 ,

$$\frac{E_s}{N_0} = 2 \frac{E_b}{N_0}.$$

Thus,

$$\boxed{\gamma_s = 2\gamma_b.}$$

In dB scale,

$$\boxed{\gamma_s(\text{dB}) = \gamma_b(\text{dB}) + 10 \log_{10} 2 = \gamma_b(\text{dB}) + 3.01 \text{ dB}.}$$

This means that for the same numerical energy per bit, the energy per symbol is 3.01 dB larger because each symbol carries two bits.

10 Part (j): Bit Error Probability and Gray Encoding

In this part, bit error probabilities were estimated for two different bit mappings. The original mapping is

$$00 \rightarrow s_1, \quad 01 \rightarrow s_2, \quad 10 \rightarrow s_3, \quad 11 \rightarrow s_4.$$

The Gray mapping is

$$00 \rightarrow s_1, \quad 01 \rightarrow s_2, \quad 11 \rightarrow s_3, \quad 10 \rightarrow s_4.$$

The receiver is still the ML receiver derived in Part (c). The constellation itself does not change; only the bit labels assigned to the constellation points change.

For each value of γ_b , the corresponding symbol SNR was found from

$$\gamma_s = 2\gamma_b.$$

The simulation results are given in Table 3.

Table 3: Bit error probabilities for original and Gray mappings.

γ_b (dB)	Original mapping BER	Gray mapping BER
-20	4.682230×10^{-1}	4.442495×10^{-1}
-16	4.480840×10^{-1}	4.110940×10^{-1}
-12	4.119475×10^{-1}	3.611195×10^{-1}
-10	3.837990×10^{-1}	3.267685×10^{-1}
-8	3.477890×10^{-1}	2.862555×10^{-1}
-6	3.016425×10^{-1}	2.393245×10^{-1}
-4	2.446675×10^{-1}	1.863190×10^{-1}
-2	1.784810×10^{-1}	1.305255×10^{-1}
0	1.116500×10^{-1}	7.862500×10^{-2}
2	5.529250×10^{-2}	3.782000×10^{-2}
4	1.853550×10^{-2}	1.253000×10^{-2}
6	3.430000×10^{-3}	2.290000×10^{-3}
8	2.630000×10^{-4}	1.815000×10^{-4}

The BER curves are shown in Fig. 7. The Gray mapping gives lower bit error probability than the original mapping. The reason is that most symbol errors at moderate and high SNR occur to nearest-neighbor constellation points. With Gray encoding, neighboring symbols differ by only one bit, so most symbol errors cause only one bit error. In the original mapping, some neighboring constellation points differ by two bits, so a nearest-neighbor symbol error can produce two bit errors.

The symbol error probability is not changed by Gray encoding because the constellation points and the ML decision regions are the same. However, the bit error probability changes because the bit labels of the detected symbols are different.

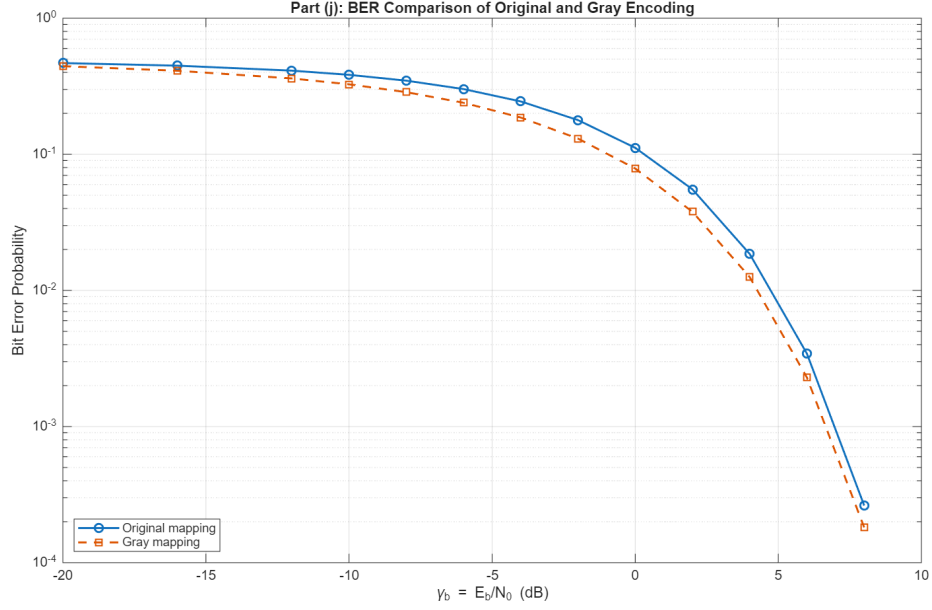


Figure 7: Bit error probability comparison for original and Gray bit mappings.

11 Conclusion

In this assignment, experiments showed that MC simulation matched the theoretical expression closely.

For unequal symbol probabilities, The simulation showed that the MAP receiver gives lower symbol error probability than the ML receiver when the symbols are not equally likely which is expected.

Finally, the bit error simulations showed that Gray encoding improves BER because nearest-neighbor symbol errors usually cause only one bit error under Gray labeling. Therefore, while Gray encoding does not change the symbol error probability, it improves the bit error probability.

Overall, these are the results of the experiments done in this assignment.